

Volume of Rectangular Prisms

Lesson 5-10

Name:

Date:

Class:



TODAY'S OBJECTIVES

Content Objective: I can find the volume of a rectangular prism, including ones with fractional edge lengths, using $\text{base area} \times \text{height}$.

Language Objective: I can explain my work using the words volume, rectangular prism, dimensions, and base area.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Dimensions

Base area



Tap any word to see its meaning and a picture.

1 The amount of space inside a three-dimensional solid is its

2 A solid with six rectangular faces is a .

3 The unit used to measure volume is .

4 The measurements of length, width, and height of a solid are its

5 The area of the bottom face of a prism, multiplied by height to find volume, is the

Write About the Math

How many gallons of water does the small tank hold?

¿Cuántos galones de agua tiene la pecera pequeña?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Volume**
Volumen

☐ **Rectangular prism**
Prisma rectangular

☐ **Cubic units**
Unidades cúbicas

☐ **Dimensions**
Dimensiones

☐ **Base area**
Área de la base

Model: You still multiply $\text{length} \times \text{width} \times \text{height}$ even when an edge is a fraction or decimal. — Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3 \text{ ft}^3$) and that a fractional edge just means multiplying with a fraction or decimal.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The tank's volume is ____ in³ because $V = \text{ } \times \text{ } \times \text{ }$. It holds about ____ gallons because $\text{ } \div 231 \approx \text{ }$.**

- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **My evidence is ____, so ____.**

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.